

Solving Absolute Value Linear Equations

TEACHER KEY | Algebra II | Unit 2 | Day 5 | TEK A2.6.E

Objective: I can solve absolute value linear equations by isolating the absolute value and splitting into two cases.

Vocabulary

Isolate — get the absolute value expression **alone** on one side before splitting.

Split into two cases — $|A| = k$ ($k > 0$) becomes $A = k$ **OR** $A = -k$.

Check — plug each solution back into the **original** equation before accepting it.

Part 1 — The Method

Step 1: Isolate the absolute value. Step 2: Split into two cases. Step 3: Solve each. Step 4: Check.

Example 1. $|2x - 3| - 4 = 3$

Isolate: $|2x - 3| = 7$

Split: $2x - 3 = 7$ OR $2x - 3 = -7$

Solve: $x = 5$ OR $x = -2$ Solution set: $\{-2, 5\}$

Part 2 — More Examples

Example 2. $|x| + 2 = 5$

Isolate: $|x| = 3$ Split/Solve: $x = 3$ or $x = -3$ (both valid)

You Try. $|3x + 6| = 12$

$3x + 6 = 12$ or $3x + 6 = -12$ $x = 2$ or $x = -6$

Part 3 — Isolate First, Then Split

Example 3. $2|x + 1| - 3 = 7$

Isolate: $2|x + 1| = 10 \rightarrow |x + 1| = 5$

Split/Solve: $x + 1 = 5$ or $x + 1 = -5$ $x = 4$ or $x = -6$

You Try. $3|x - 4| + 1 = 13$

$|x - 4| = 4$ $x = 8$ or $x = 0$

Practice

1. $|x - 4| = 9$ $x = 13$ or $x = -5$

2. $|5x| - 2 = 18$ $x = 4$ or $x = -4$

3. $4|x + 2| = 20$ $x = 3$ or $x = -7$

Exit Ticket

Solve $2|x - 5| - 1 = 9$, showing the isolate and split steps.

$|x - 5| = 5$ $x = 10$ or $x = 0$